PIPELINE AUDIT AND HEALTH REPORT

# Manufacturing: beverage bottling line

A high-speed filler, capper, labeler, fill-level check, case packer (batches of 12) and palletizer, linked by short conveyors with small buffers. The filler is the slowest machine and breaks down, so the machines downstream of it starve while the ones upstream block. This is a 30-minute run measured in bottles; use 5x playback speed.

Generated 2026-10-13 19:48 | simulated 0.5 h (warm-up 2 min) | seed 8 | 10 replications

HEALTH SCORE

**82** /100

**Grade B**

## Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 1 | Medium 1 | Low 1 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

| THROUGHPUT | COMPLETED | ARRIVALS LOST |
|---|---|---|
| 2828.6 units/h | 1392 units | 0.0% |

| AVERAGE WIP | CYCLE TIME | BOTTLENECK |
|---|---|---|
| 56.1 units | 69 s | filler |

## Executive summary

Primary bottleneck: 'filler' (96% busy, 0% down, index 96). Sources are held back up to 19% of the time. Run what-if to measure which lever recovers the most throughput.

### What to fix first

- High (filler): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'filler' (+10.9% throughput).
- Medium (filler, labeler): Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.
- Low (run length): Aim for at least 30x the cycle time so averages settle.

### Best measured improvement

Best lever: 10% faster processing at 'filler' -> +301.3 units/hour (10.9%), measured over 3 seeds with common random numbers.

## Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

| SEVERITY | AREA | WHERE | ISSUE AND RECOMMENDED ACTION | POINTS |
|---|---|---|---|---|

| SEVERITY | AREA | WHERE | ISSUE AND RECOMMENDED ACTION | POINTS |
|---|---|---|---|---|
| High | Capacity | **filler** | The bottleneck is 96% utilised (busy or down). There is no slack: any disturbance directly cuts output.<br>Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'filler' (+10.9% throughput). | -8 |
| Medium | Reliability | **filler, labeler** | Every unit passes through these stations, and they can fail. A single failure stops the whole plant.<br>Action: Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair. | -8 |
| Low | Model | **run length** | The measured window (0.5 h) is only 24x the cycle time.<br>Action: Aim for at least 30x the cycle time so averages settle. | -2 |
| Info | Capacity | **filler** | Constraint of the line: 96% utilised.<br>Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'filler' (+10.9% throughput). | |

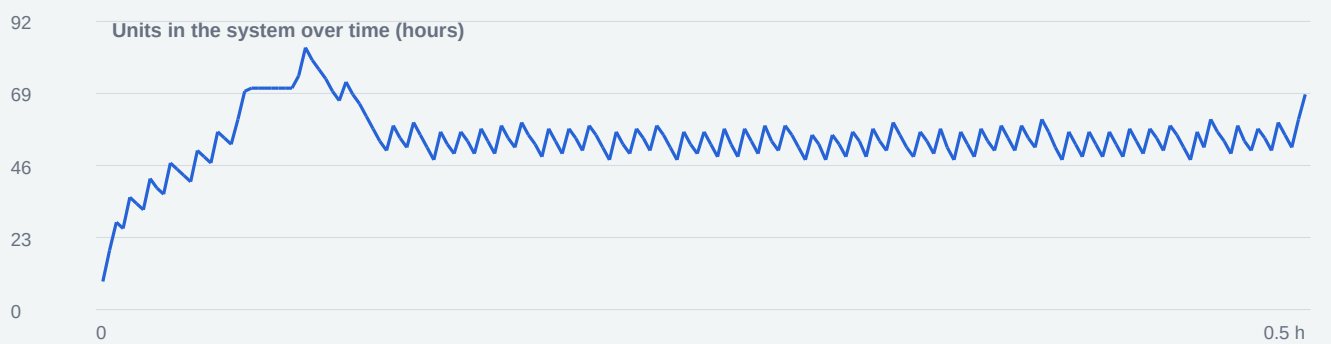
## Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

| NODE | KIND | STATE MIX | BUSY % | BLOCKED % | DOWN % | AVG / MAX WIP |
|---|---|---|---|---|---|---|
| **bottles** | source | - | - | - | - | - |
| **filler**<br>bottleneck | machine | | 95.6 | 4.4 | 0.0 | 30.8 / 31 |
| **conv_fill** | delay | | 93.0 | 7.0 | 0.0 | 3.8 / 12 |
| **capper** | machine | | 68.0 | 6.0 | 0.0 | 1.6 / 11 |
| **labeler** | machine | | 71.8 | 0.0 | 7.2 | 2.1 / 11 |
| **check** | quality_gate | | 23.9 | 0.0 | 0.0 | 0.2 / 1 |
| **packer** | batcher | | 39.4 | 0.0 | 0.0 | 10.2 / 19 |
| **palletizer** | machine | | 59.1 | 0.0 | 0.0 | 7.1 / 12 |
| **pallets** | sink | - | - | - | - | - |
| **rejects** | sink | - | - | - | - | - |

### Work in process over time




## Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

| METRIC | MEAN | 95% CONFIDENCE | STD DEV | RELATIVE |
|---|---|---|---|---|
| **Throughput** | 2831.14 units/h | +- 53.06 | 74.18 | 1.9% |
| **Average WIP** | 54.12 units | +- 0.95 | 1.32 | 1.7% |
| **Cycle time** | 66.80 s | +- 1.19 | 1.66 | 1.8% |
| **Arrivals lost** | 0.00 % | +- 0.00 | 0.00 | 0.0% |

**No replication deadlocked.**

### Failure model

| NODE | MTBF | MTTR | AVAILABILITY | OBSERVED DOWN | FAILURES |
|---|---|---|---|---|---|
| **filler** | 15 min | 0.7 min | 95.7% | 0.0% | 0 |
| **labeler** | 20 min | 0.5 min | 97.6% | 7.2% | 2 |

### Single points of failure

All units must pass through: filler, capper, labeler, check. If one of these stops, output stops.

## Model validity checks

| CHECK | RESULT | DETAIL |
|---|---|---|
| **Unit conservation** | PASS | created 1475 = completed 1392 + scrapped 18 + lost 0 + in system 65 |
| **Little's Law** | PASS | L = 56.07, lambda x W = 55.20 (gap 1.5%) |
| **Run length** | WARN | window = 24 x cycle time (30x or more recommended) |
| **Warm-up** | PASS | first 2 min excluded from statistics |
| **Deadlock-free** | PASS | reference run completed |

## What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 2774.1 units/h.

| NODE | CHANGE | UNITS/H | CHANGE | CHANGE % |
|---|---|---|---|---|
| **filler** | 10% faster processing | 3075.3 | **+301.3** | +10.9% |
| **filler** | 20% shorter repairs | 2803.2 | **+29.1** | +1.0% |
| **labeler** | 20% shorter repairs | 2779.6 | **+5.5** | +0.2% |
| **labeler** | +2 buffer slots | 2775.6 | **+1.5** | +0.1% |
| **capper** | +2 buffer slots | 2775.6 | **+1.5** | +0.1% |
| **labeler** | 10% faster processing | 2775.1 | **+1.0** | +0.0% |
| **filler** | +2 buffer slots | 2774.1 | **+0.0** | +0.0% |
| **capper** | 10% faster processing | 2774.1 | **+0.0** | +0.0% |

**Best lever: 10% faster processing at 'filler' -> +301.3 units/hour (10.9%), measured over 3 seeds with common random numbers.**

## Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

| NODE | TYPE | EFFECTIVE PARAMETERS |
|---|---|---|
| **bottles** | source | interarrival=1.00, on_blocked=wait |
| **filler** | machine | capacity=30, mtbf=900, mttr=40, process=normal(1.20,0.10) |
| **conv_fill** | delay | capacity=12, transit=4 |
| **capper** | machine | capacity=10, process=normal(0.85,0.05) |
| **labeler** | machine | capacity=10, mtbf=1200, mttr=30, process=normal(0.90,0.08) |
| **check** | quality_gate | capacity=8, max_reworks=0, pass_to=packer, process=0.30, scrap_to=rejects, threshold=0.02 |
| **packer** | batcher | batch_size=12, capacity=24, process=6 |
| **palletizer** | machine | capacity=4, process=9 |
| **pallets** | sink | (defaults) |
| **rejects** | sink | counts_as=scrapped |

## Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

| SEVERITY | WHERE | FINDING | POINTS |
|---|---|---|---|
| High | **filler** | The bottleneck is 96% utilised (busy or down). There is no slack: any disturbance directly cuts output. | -8 |
| Medium | **filler, labeler** | Every unit passes through these stations, and they can fail. A single failure stops the whole plant. | -8 |
| Low | **run length** | The measured window (0.5 h) is only 24x the cycle time. | -2 |
| | **Total deductions** | | **-18** |
| | **Health score** | | **82** |

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.